

The Economy

Seminar 3: Resources and Economic Decisions

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Consumer preferences: Assumptions

We impose three assumptions about consumer preferences: preferences are complete, transitive and non-satiated.

Key assumptions.

- ① **completeness:** for any two bundles A and B , a consumer can establish a preference ordering. That is, for any comparison of bundles, they will choose one and only one of the following:
 - $A \succ B \rightarrow A$ is strictly preferred to B
 - $B \succ A \rightarrow B$ is strictly preferred to A
 - $A \sim B \rightarrow$ indifferent between A and B
- ② **transitivity:** for any consumer if $A \succsim B$ and $B \succsim C$, then it must be that $A \succsim C$.
- ③ **non-satiation:** if bundle A has more of every good compared to bundle B , then $A \succ B$.

Given these three assumptions are obeyed, we can always define a utility function. Any utility function that satisfies these assumptions **cannot have indifference curves that cross**.

Comment on completeness and transitivity

Example

Imagine that you have to choose among three artists: ♦ Daft Punk ♦ Taylor Swift ♦ BTS. Now consider the following questions

- Which is better? Daft Punk or Taylor Swift.
- Which is better? Taylor Swift or BTS.
- Which is better? Daft Punk or BTS.

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-
- Were you able to respond all questions? Then your preferences are **complete**.
 - Did your preferences create a cycle? Then your preferences are **transitive**.

Indifference curves: Properties 1 & 2

- 1 There is an indifference curve for each bundle of goods.
- 2 Always prefer higher indifference curves → due to the **non-satiation assumption** of consumer preferences.

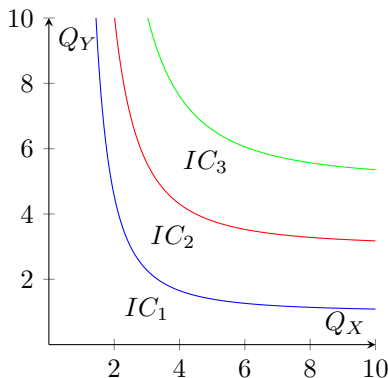
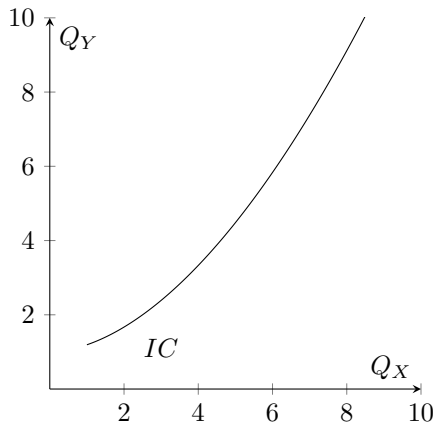


Figure 1: Three indifference curves.

Indifference curves: Property 3

- Indifference curves are always **downwards sloping**.

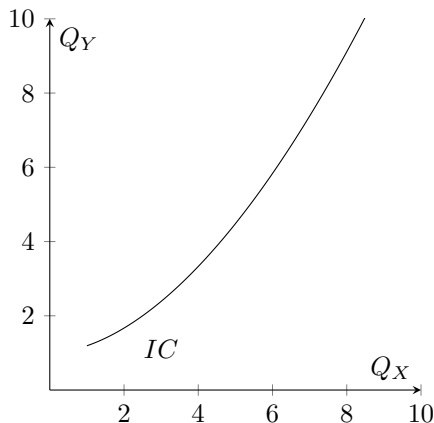


Why?

Figure 2: An **upwards** sloping indifference curve.

Indifference curves: Property 3

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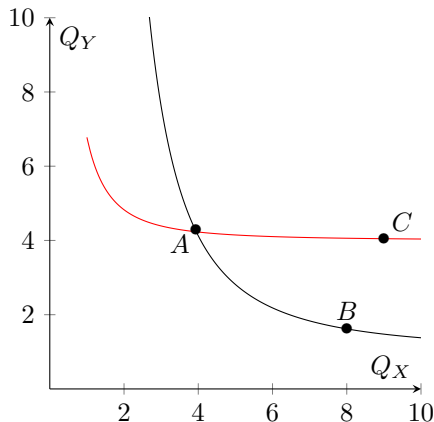
Why?

violates the
non-satiation
assumption!

Figure 2: An **upwards** sloping indifference curve.

Indifference curves: Property 4

- ④ Indifference curves **cannot intersect** with each other.



Why?

Figure 3: Intersecting indifference curves.

Indifference curves: Property 4

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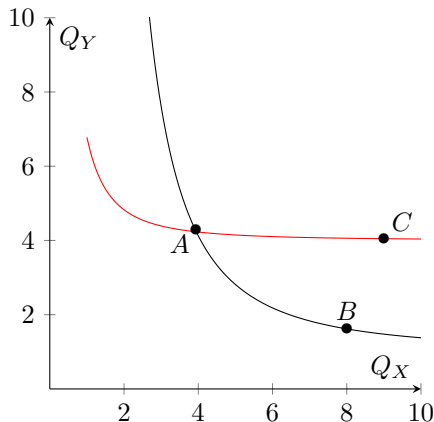
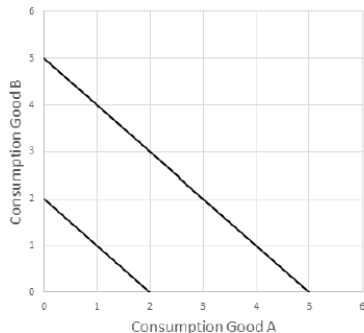


Figure 3: Intersecting indifference curves.

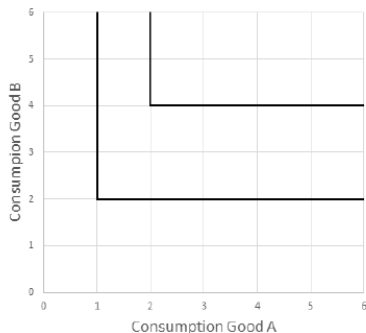
Why?

violates both the
non-satiation and the
transitivity
assumptions!

PS3 Exercise 2. Indifference curves



(a) Figure 1



(b) Figure 2

- 1 How would you characterize the relationship between goods A and B in case 1? In case 2?
- 2 For each of the graphs, give an example of two goods that might correspond to such indifference curves.

Utility maximization

Basic model of individual choice:

- A decision-maker (DM) must choose one **alternative** x from a set X .
- Chooses to maximize a utility function u .
- u specifies how much utility DM gets from each alternative:

$$u : X \rightarrow \mathbb{R}$$

Example: DM chooses whether to eat an apple or a pear.

$$X = \{apple, pear\}.$$

Utility function might say $u(apple) = 7$, $u(pear) = 12$
→ apple gives 7 units of pleasure, pear gives 12 units of pleasure.

This is **not** the standard way economists think about utility.

Utility levels and preference interpretation

Utility represents an individual's choices.

- Individual choices are **primitive data** that economists can observe.
- Choices are taken to **reveal** individual's preferences.
- Utility is a **convenient mathematical construction** for modeling choices and preferences.

$“u(\text{apple}) = 7, u(\text{pear}) = 12” \rightarrow$ individual prefers pears to apples.

$“u(\text{apple}) = 2, u(\text{pear}) = 15” \rightarrow$ individual prefers pears to apples.

Preference and utility

Now that know how to infer preferences from choice, next step is representing preferences with a utility function.

A utility function $u : X \rightarrow \mathbb{R}$ **represents** preference relation $\succsim$ if, for all x, y

$$x \succsim y \Leftrightarrow u(x) \geq u(y)$$

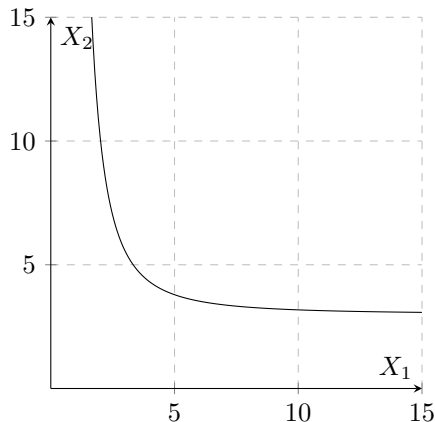
pear $\succsim$ *apple* are represented by **both**:

$u(\text{apple}) = 7, u(\text{pear}) = 12$ and
 $u(\text{apple}) = 2, u(\text{pear}) = 15$.

- if u represents $\succsim$, so does any strictly increasing transformation of u .
- representing a given preference relation is an **ordinal property**.
- the numerical values of utility are **cardinal properties**.

Marginal Rate of Substitution

Given what we know about utility, we can now link it to preference maps using the **marginal rate of substitution**.



Graphically, MRS is **the slope of the indifference curve**. It is the rate at which you're willing to **trade off one good for the other**. This is represented by:

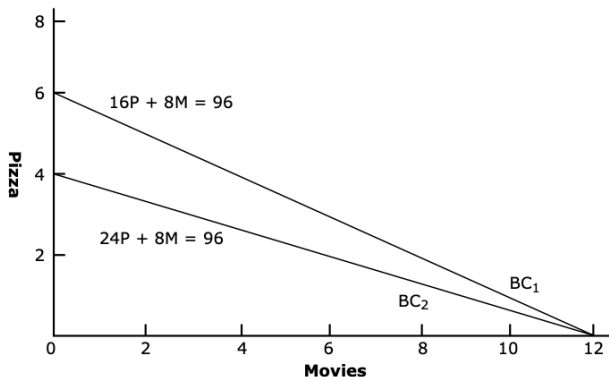
$$MRS_{ij}(\mathbf{x}) = -\frac{\partial u(\mathbf{x})/\partial x_i}{\partial u(\mathbf{x})/\partial x_j}$$

Figure 4: MRS_{12} shown graphically.

Budget constraints

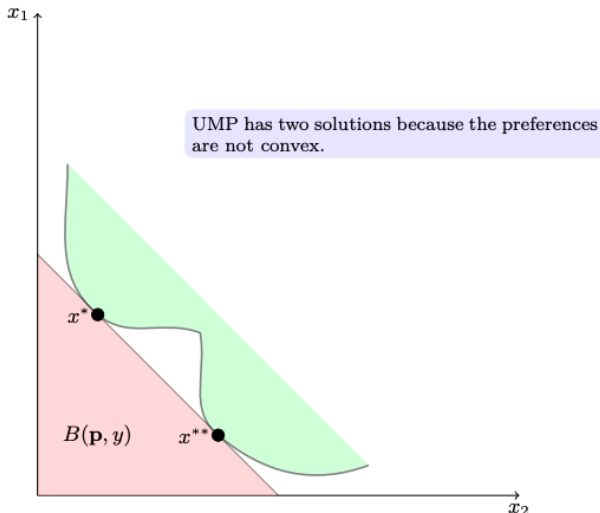
Budget constraint over two goods X and Y is defined as:

$$I = p_X X + p_Y Y$$

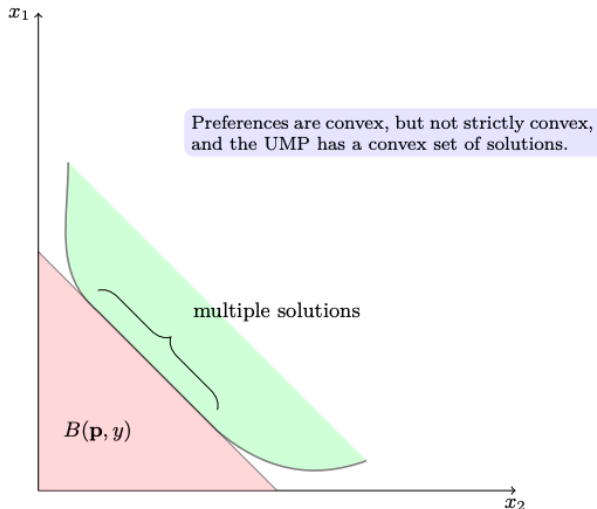


Budget constraints - Potential scenarios

Recall that we need **convex sets** of preferences.



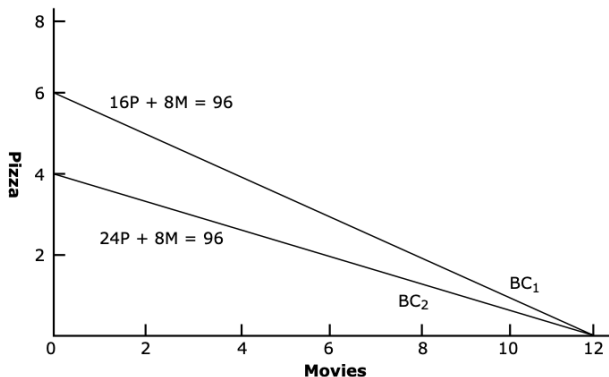
Budget constraints - Potential scenarios



Budget constraints - Increase in prices

An increase in price has two effects:

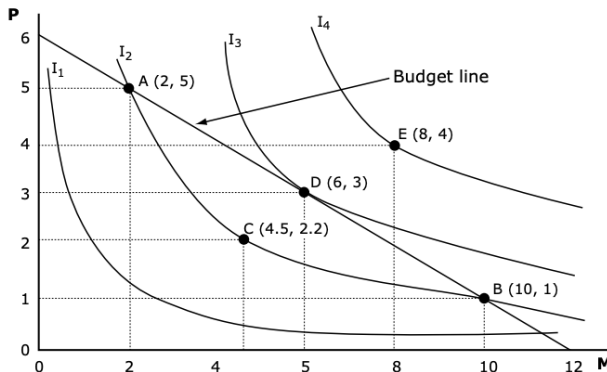
- **income effect** - it makes the consumer relatively poorer.
- **substitution effect** - it also makes this specific good less attractive relative to alternatives.



Budget constraints - MRT

The rate at which you can transform one good into the other is called the **marginal rate of transformation**. This is equal to the MRS at the with the optimal bundle.

$$MRS_{xy} = MRT_{xy} = -\frac{p_X}{p_Y}$$



PS3 Exercise 1. Consumption or leisure (Part 1)

We consider an individual who can freely choose his daily working time. His hourly wage is fixed at 20 euros.

- ① What is the opportunity cost for this agent to consume a good worth 30 euros in hours?
- ② Graphically represent the set of combinations of number of hours of leisure / consumption that this agent can choose. Put the number of rest hours on the x-axis. We suppose that this number of hours is at least equal to 12.

We know that this agent is indifferent between:

- Working 5 hours a day for a daily wage of 130 euros
- Working 7 hours a day for a daily wage of 140 euros
- Working 9 hours a day for a daily wage of 210 euros

As well as between:

- Working 5 hours a day for a daily wage of 150 euros
- Working 7 hours a day for a daily wage of 220 euros
- Working 9 hours a day for a daily wage of 300 euros

PS3 Exercise 1. Consumption or leisure (Part 2)

- ③ Draw the two indifference curves corresponding to this information on the previous graph.
- ④ What is the optimal number of working hours for this agent?
- ⑤ What would be his optimal number of working hours if his wage were to increase to 30 euros per hour ? Which effect would then dominate between the substitution effect and the income effect?